Experimental Report: Multilingual Word Embeddings for Indian Languages

Alok Sahni (G25AIT2138)
PGDAI, SAIDE
Indian Institute of Technology Jodhpur
Jodhpur, India
g25ait2138@iitj.ac.in

***Abstract*—This report trains and evaluates 300-dimensional word embeddings for seven Indian languages—Gujarati, Hindi, Marathi, Punjabi, Tamil, Telugu and Urdu—spanning six writing systems. One FastText Skip-gram model per language is trained from random initialisation on the IndicCorpV2-5k corpus only, with character 3–6-grams for out-of-vocabulary (OOV) words and a fallback vector so that no word pair is discarded. Against the human similarity judgements of Akhtar et al., the models reach a macro-average Spearman correlation of $\rho = 0.3244$, from 0.2414 (Marathi) to 0.4484 (Gujarati), with every correlation significant at $p < 0.05$. An ablation over all seven languages shows that each change to the main configuration lowers the macro score; a Word2Vec baseline without subwords reaches only 0.2392; and retraining with three seeds gives 0.3275 $\pm$ 0.0060. Training is deterministic, so every reported number is exactly reproducible.**

***Index Terms*—word embeddings, FastText, Indian languages, subword models, out-of-vocabulary, word similarity**

I. Introduction

Distributional word embeddings map words to dense vectors whose geometry reflects co-occurrence statistics, so that words used in similar contexts receive similar vectors [2]. For Indian languages, rich inflectional and agglutinative morphology produces many rare surface forms, and several scripts with combining marks must be handled correctly. Subword models such as FastText [1] represent a word as the sum of its character n-gram vectors, which also yields vectors for words never seen in training.

This work builds seven monolingual 300-dimensional FastText models from scratch on the IndicCorpV2-5k subset [3] and evaluates them intrinsically on the word-similarity datasets of Akhtar et al. [4].

A. Objectives and Scope

The study has three aims: (i) train embeddings from first principles under a fixed data budget, (ii) design an OOV strategy under which no evaluation pair is discarded, and (iii) measure, on all seven languages, how sensitive the models are to n-gram range, epoch budget, vocabulary threshold, subwords and random seed. Every benchmark word, including unseen inflections and blank fields, must receive a vector, and the exported arrays must keep the original pair order with shape $(N, 2, 300)$. The benchmark is used for evaluation only; its human scores never enter training or model selection.

TABLE I
Learning Objectives and Where They Were Met

Objective	How it was met
Train distributional embeddings from first principles	Seven FastText Skip-gram models from random initialisation at exactly 300 dimensions, no pre-trained vectors (Sec. II-C)
Understand subword representation for rich morphology	Character 3–6-grams as the OOV mechanism, measured against 2–5-grams and a Word2Vec baseline (Sec. II-D, V)
Design an OOV strategy instead of discarding cases	Subword composition plus a frequency-weighted fallback vector for any input (Sec. II-D)
Apply correct intrinsic evaluation	Cosine similarity, Spearman correlation with p-values, macro-averaging and rank-based error analysis (Sec. II-F, VI)
Process multilingual Unicode text	NFC normalisation shared by training and evaluation across six scripts (Sec. II-B)
Handle imperfect real-world data	Mixed delimiters, BOMs, missing scores and a blank field handled while keeping every row in place (Sec. II-E)
Investigate hyperparameter sensitivity	Ablation on all seven languages, a Word2Vec baseline and three random seeds (Sec. V)
Interpret behaviour and state limitations	Rank-based diagnosis of correct and incorrect predictions; explicit limitations (Sec. VI, VII)

B. Learning Objectives

Table I pairs each learning objective with the part of this work where it was exercised.

II. Methodology

A. Pipeline

Fig. 1 shows the pipeline. Benchmark words pass through the same Unicode normalisation as training text, and human scores reach only the evaluation stage.

B. Corpus, Normalisation and Tokenisation

The dataset `rony000013/indiccorpV2-5k` holds 118,243 records across 24 languages, with fields `language`, `code` and `text`. Records are selected by their `code`, giving exactly 5,000 per benchmark language. Text is normalised to Unicode NFC, split on whitespace, and punctuation, including the sentence-final danda, is stripped from each token; no case-folding or stemming is applied. For example, the Hindi sentence *acchā chātra vidyālay mẽ paṛhtā hai* ("a good student studies at school") yields six tokens. Table II gives the resulting corpus sizes; with `min_count`=1 every token type enters the vocabulary.

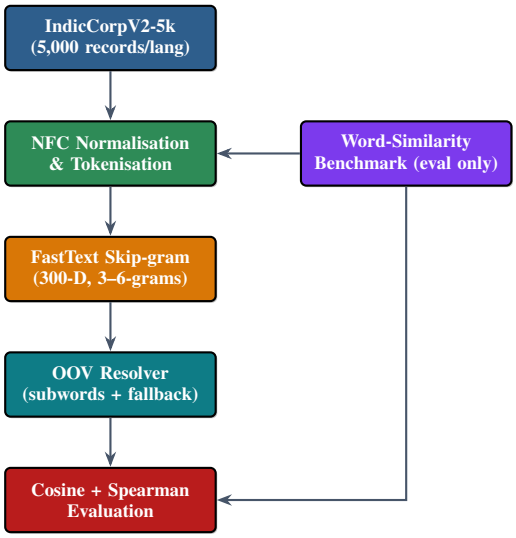


Fig. 1. Training and evaluation pipeline. Benchmark words share the preprocessing path; human scores reach only the evaluation stage.

TABLE II
TRAINING CORPUS PER LANGUAGE (5,000 RECORDS EACH)

Language	Tokens	Vocabulary
Gujarati	203,449	41,355
Hindi	284,598	27,476
Marathi	206,264	47,657
Punjabi	272,700	28,777
Tamil	168,026	58,128
Telugu	172,252	49,794
Urdu	362,086	37,758

TABLE III
MAIN HYPERPARAMETERS (IDENTICAL FOR ALL LANGUAGES)

Parameter	Value	Parameter	Value
Architecture	Skip-gram	Negative samples	5
Vector size	300	Char n-grams	3–6
Window	5	n-gram buckets	2,000,000
Min. count	1	Subsampling	10^{-3}
Epochs	25	Learning rate	0.025
Seed	42	Workers	1

C. Model and Main Hyperparameters

One independent Gensim [7] FastText model is trained per language from random initialisation, with the identical configuration in Table III. FastText was chosen for its character n-gram composition, which makes principled OOV inference possible.

D. Out-of-Vocabulary Handling

Every input receives a finite, non-zero 300-dimensional vector: (i) in-vocabulary words use their learned vector; (ii) unseen words are composed from their learned character 3–6-gram vectors; (iii) a word that yields no usable vector—a blank field or a near-zero subword vector—falls back to the frequency-weighted mean vocabulary vector

$$\mathbf{v}_{\text{fb}} = \frac{\sum_{w \in V} c_w \, \mathbf{v}_w}{\sum_{w \in V} c_w}, \tag{1}$$

where V is the vocabulary, c_w the training count of w and $\mathbf{v}_w$ its vector. In this run the fallback is reached only by the single Tamil row with a blank word; no word pair is discarded.

TABLE IV
PAIR COUNTS: ROWS EXCLUDED FROM THE CORRELATION ONLY

Language	N	Scored	Excl.	Reason
Marathi	253	104	149	No human score
Tamil	97	96	1	Blank word field
Other five	752	752	0	—

E. Benchmark Integrity and Pair Counts

The Hindi file is comma-separated and the others tab-separated, so the delimiter is detected per line; files are read as `utf-8-sig` to strip byte-order marks; and missing scores become NaN. All N rows of each file are embedded and exported in their original order, but only rows with two non-blank words and a finite human score enter the correlation. Table IV explains why the number of scored pairs differs from N.

Evaluation-only guarantee: human similarity scores are used only to compute the reported correlations; they never enter training, gradients, model selection or early stopping.

F. Evaluation Metric

For each pair (w_1, w_2), similarity is the cosine $s = \mathbf{v}_1^\top \mathbf{v}_2 / (\|\mathbf{v}_1\| \, \|\mathbf{v}_2\|)$. Per language, Spearman's rank correlation [5] between model scores s_i and human scores h_i is

$$\rho = \frac{\operatorname{cov}\big(R(h), R(s)\big)}{\sigma_{R(h)} \, \sigma_{R(s)}}, \tag{2}$$

where $R(\cdot)$ denotes ranks. The headline metric is the unweighted mean of ρ over the seven languages.

G. Implementation and Reproducibility

The pipeline is a single Jupyter notebook. Training uses one worker thread and a fixed CRC-32 hash in place of Python's randomised `hash()`, which makes training bit-for-bit deterministic: re-running the notebook reproduces every number in this report. For each language the notebook exports `<Lang>.npy`, a float32 array of shape $(N, 2, 300)$ in original pair order; `<Lang>_vocab.json`, mapping every benchmark word to its `[row, column]` positions with its in-vocabulary status; and a per-pair prediction table. Before packaging, a validator checks array shape, dtype, finiteness and non-zero norms, the stored pair order against the benchmark, and OOV inference after reloading each model. An inference interface (`get_word_vector`, `embed_words`, `embed_pairs`, `predict_similarity`) loads the saved models to embed arbitrary words for the private evaluation;

TABLE V
PER-LANGUAGE RESULTS ON THE PUBLIC BENCHMARK

Language	N	Scored	ρ	p-value	Cov.
Gujarati	163	163	**0.4484**	<0.0001	89.6%
Hindi	235	235	0.3009	<0.0001	87.7%
Marathi	253	104	0.2414	0.0136	79.8%
Punjabi	143	143	0.4043	<0.0001	89.3%
Tamil	97	96	0.3204	0.0015	82.4%
Telugu	111	111	0.2453	0.0095	72.5%
Urdu	100	100	0.3101	0.0017	82.9%
Macro / total	**1,102**	**952**	**0.3244**	—	**83.9%**

because each model occupies about 2.4 GB (two million n-gram buckets), the models are not shipped in the archive, and the interface retrains any missing model deterministically.

H. Data Sources and External-Resource Disclosure

Training data: `rony000013/indiccorpV2-5k` only [3]. No pre-trained models or embeddings and no external training corpora are used. *Evaluation data:* the word-similarity datasets of Akhtar et al. [4], used only to compute the reported correlations. *Tools:* Python, Gensim (FastText, Word2Vec), NumPy, SciPy, pandas, Matplotlib and scikit-learn (t-SNE). All links are listed under Resources at the end of this report.

III. T-SNE VISUALIZATION

Fig. 2 shows 100 word samples per language: the 100 most frequent vocabulary words of each model, projected from 300 to 2 dimensions with t-SNE [6] (perplexity 25, PCA initialisation, seed 42). These are mostly function words and frequent content words; they spread fairly evenly with a few tight clusters, typically of inflectional variants of one stem, which share many character n-grams. Dimensionality reduction is used only for this figure; all evaluation uses the full 300-dimensional vectors. Points carry numeric IDs, mapped back to words in a CSV file per language, because the plotting environment lacks fonts for all six scripts.

IV. RESULTS

Cosine similarity was computed for all 1,102 benchmark pairs and Spearman's ρ (2) on the 952 scored pairs (Table V, Fig. 3). The macro-average is $\rho = 0.3244$. Gujarati is best ($\rho = 0.4484$) and Marathi weakest ($\rho = 0.2414$); all seven correlations are significant at $p < 0.05$.

A. Vocabulary Coverage and OOV Statistics

Table VI reports coverage over the unique words of each benchmark. Overall, 1,279 of 1,524 benchmark words (83.9%) occur in training; the remaining 245 are represented by subword composition. Coverage ranges from 72.5% (Telugu) to 89.6% (Gujarati) and tracks quality closely: across languages, the rank correlation between coverage and ρ is 0.82 ($p = 0.023$). The two highest-coverage languages, Gujarati and Punjabi, are also the two best, while Telugu and Marathi have the lowest coverage and the lowest correlations.

TABLE VI
VOCABULARY COVERAGE AND OOV STATISTICS

Language	Unique	In vocab	OOV	OOV rate
Gujarati	221	198	23	10.4%
Hindi	318	279	39	12.3%
Marathi	337	269	68	20.2%
Punjabi	206	184	22	10.7%
Tamil	136	112	24	17.6%
Telugu	160	116	44	27.5%
Urdu	146	121	25	17.1%
Total	**1,524**	**1,279**	**245**	**16.1%**

TABLE VII
OOV INFERENCE ON HINDI INPUTS

Pair (romanised)	Cosine	Case
vidyālay / *skūl*	0.5637	Synonyms (school), both in vocabulary
vidyālayō / *vidyālay*	0.9039	Plural and base form, both in vocabulary
malformed string / *vidyālay*	0.5708	Garbled input, resolved via subwords

Table VII shows the resolver on Hindi inputs of increasing difficulty after training.

V. EXPERIMENTS

All experiments train from scratch on IndicCorpV2 only, at 300 dimensions, on all seven languages; benchmark scores are only read off afterwards.

A. Ablation

Each variant changes one setting of the main configuration (Table IX, Fig. 4). Every change lowers the macro score: shorter 2–5-grams cost the most (-0.0567), followed by 15 instead of 25 epochs (-0.0398) and `min_count`=2 (-0.0177). Longer n-grams capture morpheme-sized units, more epochs give the small corpus more passes, and `min_count`=1 keeps rare words—raising coverage—at the cost of noisier rare-word vectors. Individual languages deviate: Hindi scores higher with 2–5-grams (0.3226), Tamil with 15 epochs (0.3383) and Telugu with `min_count`=2 (0.2852). These gains are several times the per-language seed standard deviations (Sec. V-C), so per-language settings would likely help. A single configuration was kept because it maximises the macro score and because choosing settings per language on the public benchmark, which is also the development set, risks overfitting to it.

B. Word2Vec Baseline

A Word2Vec Skip-gram model [2] with identical settings but no subwords isolates the contribution of character n-grams (Table X, Fig. 5); its OOV words all receive the fallback vector. FastText is better in six of seven languages and by +0.0852 macro (36% relative). The largest gains are in Punjabi (+0.177), Urdu (+0.116), Marathi (+0.111) and Tamil (+0.105); only Telugu is marginally lower (-0.011). Subwords help both OOV words and rare in-vocabulary words, whose vectors are stabilised by n-grams shared with frequent words.

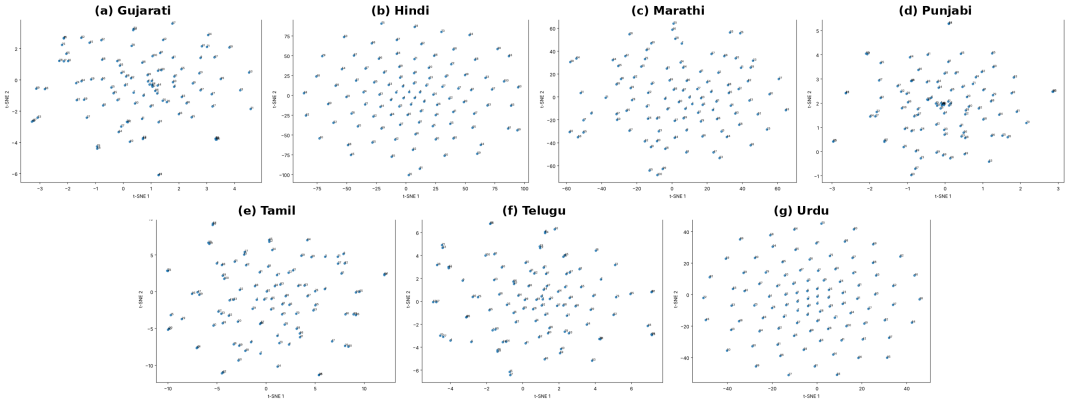


Fig. 2. t-SNE projections of the 100 most frequent words per language. Points carry numeric IDs mapped to words in per-language CSV files.

TABLE VIII
RANK-BASED ERROR ANALYSIS: CORRECTLY AND INCORRECTLY PREDICTED PAIRS (ROMANISED, WITH ENGLISH GLOSS)

Lang.	Pair (gloss)	Human	Cosine	Human rank	Model rank	Rank error
Correctly predicted (smallest rank error)						
Guj.	*vāgh* / *vāgh* (tiger)	10.00	1.000	1	1	0
Hin.	*kampyūṭar* / *sôfṭveyar* (computer / software)	8.80	0.736	4	4	0
Mar.	*vāgh* / *vāgh* (tiger)	10.00	1.000	1	1	0
Pun.	*ṭāīgar* / *jaguār* (tiger / jaguar)	8.38	0.692	10	10	0
Tam.	*roṭṭi* / *veṇṇey* (bread / butter)	7.50	0.440	32	32	0
Incorrectly predicted (largest rank error)						
Guj.	*vilamb* / *jātivād* (delay / racism)	0.00	0.594	159.5	34	125.5
Hin.	*samācār* / *ripōrṭ* (news / report)	7.80	0.208	31	224	193
Mar.	*ṭeliphon* / *saṃvād* (telephone / communication)	9.00	0.206	6	97	91
Pun.	*lobasṭar* / *vāīn* (lobster / wine)	1.00	0.630	141.5	22	119.5
Tam.	*cakōtarar* / *tuṟavi* (brother / monk)	1.12	0.740	88	8	80

TABLE IX
ABLATION ON ALL SEVEN LANGUAGES (SPEARMAN ρ; BEST PER COLUMN IN BOLD)

Experiment	Gujarati	Hindi	Marathi	Punjabi	Tamil	Telugu	Urdu	Macro
E1 main (3–6, 25 ep., mc 1)	**0.4484**	0.3009	**0.2414**	**0.4043**	0.3204	0.2453	**0.3101**	**0.3244**
E2 n-grams 2–5	0.3964	**0.3226**	0.1743	0.2818	0.2421	0.2331	0.2239	0.2677
E3 15 epochs	0.3622	0.2313	0.1927	0.3907	**0.3383**	0.2396	0.2371	0.2846
E4 min_count 2	0.4284	0.2726	0.2198	0.3520	0.2873	**0.2852**	0.3016	0.3067

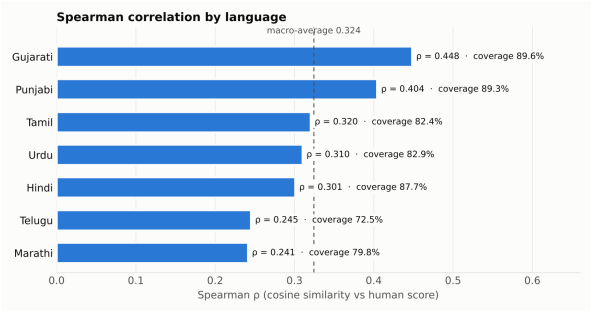


Fig. 3. Spearman correlation by language with vocabulary coverage.

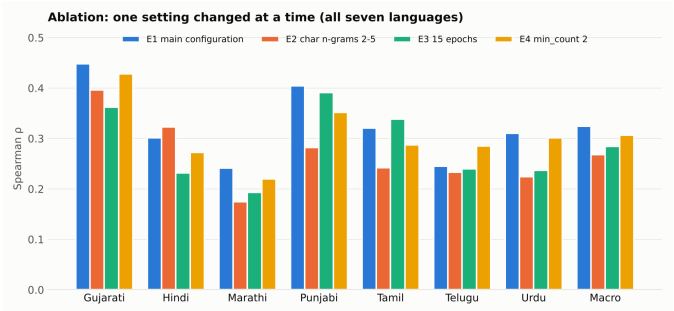


Fig. 4. Ablation: Spearman ρ per language for each one-setting change.

C. Seed Robustness

The main configuration was retrained with seeds 1, 7, 42 (Table XI). The macro score is 0.3275 $\pm$ 0.0060 (mean $\pm$ standard deviation), so the headline result is stable; individual languages vary more, which is expected with 96–235 scored

TABLE X
WORD2VEC BASELINE VERSUS FASTTEXT (SPEARMAN ρ)

Language	Word2Vec	FastText	Gain
Gujarati	0.3966	0.4484	+0.0518
Hindi	0.2548	0.3009	+0.0462
Marathi	0.1307	0.2414	+0.1106
Punjabi	0.2270	0.4043	+0.1772
Tamil	0.2154	0.3204	+0.1051
Telugu	0.2560	0.2453	-0.0107
Urdu	0.1937	0.3101	+0.1165
Macro	**0.2392**	**0.3244**	**+0.0852**

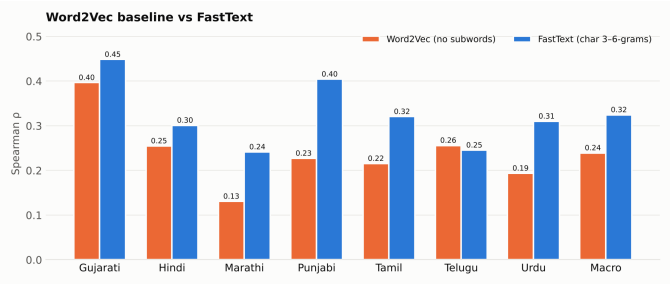


Fig. 5. Word2Vec baseline (no subwords) versus FastText.

TABLE XI
SEED ROBUSTNESS OF THE MAIN CONFIGURATION (SPEARMAN ρ)

Language	Seed 1	Seed 7	Seed 42	Mean $\pm$ SD
Gujarati	0.4481	0.4533	0.4484	0.4499 $\pm$ 0.0029
Hindi	0.3015	0.2920	0.3009	0.2981 $\pm$ 0.0053
Marathi	0.2562	0.2308	0.2414	0.2428 $\pm$ 0.0128
Punjabi	0.3954	0.3839	0.4043	0.3945 $\pm$ 0.0102
Tamil	0.3166	0.3250	0.3204	0.3207 $\pm$ 0.0042
Telugu	0.2716	0.2706	0.2453	0.2625 $\pm$ 0.0149
Urdu	0.3515	0.3098	0.3101	0.3238 $\pm$ 0.0240
Macro	**0.3344**	**0.3236**	**0.3244**	**0.3275 $\pm$ 0.0060**

pairs per language.

VI. ANALYSIS

Human scores (1–10) and cosines ($[-1, 1]$) are on different scales, so the analysis is rank-based: within each language both are ranked and the absolute rank error $|R(h) - R(s)|$ is computed. Table VIII lists ten word pairs: for each language, the pair with the smallest rank error (correctly predicted) and the pair with the largest (incorrectly predicted).

A. Correctly Predicted Similarities

Two of the five correct cases are identity pairs (Gujarati and Marathi *tiger*/*tiger*), which mainly confirm the pipeline. The other three are genuine semantic matches ranked exactly as annotators ranked them: Hindi *computer*/*software* (human 8.80, cosine 0.736, both rank 4), Punjabi *tiger*/*jaguar* (rank 10 for both) and Tamil *bread*/*butter* (rank 32). Such pairs appear in the same contexts—technology, wildlife, food—which is exactly what a Skip-gram model learns.

B. Incorrectly Predicted Similarities

The failures fall into two groups. *Under-estimated* pairs are near-synonyms that rarely share contexts: Hindi *samācār*/*ripōrṭ* ("news"/"report", human 7.80, cosine 0.208) pairs a native word with an English loan used in different registers, and Marathi *telephone*/*communication* relies on world knowledge rather than co-occurrence. *Over-estimated* pairs are unrelated words that the model places close together: Gujarati *delay*/*racism* (human 0.00, cosine 0.594), Punjabi *lobster*/*wine* (human 1.00, cosine 0.630) and Tamil *brother*/*monk* (human 1.12, cosine 0.740). These are rare words that probably co-occur in a few shared topical contexts (for example food, or religious and family texts), and with `min_count`=1 their vectors rest on very little evidence. Annotators rate similarity, not topical relatedness, so such pairs are penalised. This is the price of the higher coverage that `min_count`=1 buys, and the reason E4 helps Telugu.

VII. DISCUSSION

A. Design Choices

FastText was chosen over Word2Vec and GloVe because the no-discard requirement needs vectors for unseen words; the Word2Vec baseline confirms the choice (0.2392 versus 0.3244 macro). Seven monolingual models were trained instead of one multilingual model because the benchmark evaluates each language in isolation, and separate models avoid vocabulary interference between scripts. The final configuration—3–6-grams, 25 epochs, `min_count`=1—is supported by the ablation on all seven languages. Training is single-threaded with a fixed hash so that every reported number is exactly reproducible.

B. Limitations

Corpus size dominates: 5,000 records per language (168k–362k tokens) is a small sample, so rare-word vectors are noisy, as the over-estimated pairs show. The benchmarks hold 96–235 scored pairs, so per-language correlations carry wide confidence intervals. Static embeddings give one vector per word form and cannot resolve polysemy. The configuration was compared on the public benchmark, which is also the development set, so the public scores may be slightly optimistic for the private test set. Finally, the 2.4 GB models are regenerated rather than shipped.

C. Future Work

In order of expected value: train on more IndicCorpV2 text; tune settings per language on held-out data with several seeds; add a GloVe baseline under the same harness; and evaluate on a downstream task, since intrinsic similarity is only a proxy for usefulness.

D. Conclusion

Seven monolingual 300-dimensional FastText models trained from scratch on IndicCorpV2 reach a macro-average Spearman correlation of 0.3244 (0.3275 $\pm$ 0.0060 over three seeds), with all seven languages significant and all 1,102

benchmark pairs represented in the exported arrays. Character n-grams are the most important ingredient: they add +0.085 over Word2Vec, and shortening them to 2–5 costs 0.057. Vocabulary coverage closely tracks quality in this setting, while the main remaining errors come from rare words learned from too few contexts.

VIII. References

[1] P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov, "Enriching word vectors with subword information," *Trans. Assoc. Comput. Linguist.*, vol. 5, pp. 135–146, 2017.

[2] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," arXiv:1301.3781, 2013.

[3] S. Doddapaneni *et al.*, "Towards leaving no Indic language behind: Building monolingual corpora, benchmark and models for Indic languages," in *Proc. 61st Annu. Meeting Assoc. Comput. Linguist. (ACL)*, 2023, pp. 12402–12426.

[4] S. S. Akhtar, A. Gupta, A. Vajpayee, A. Srivastava, and M. Shrivastava, "Word similarity datasets for Indian languages: Annotation and baseline systems," in *Proc. 11th Linguistic Annotation Workshop*, 2017, pp. 91–94.

[5] C. Spearman, "The proof and measurement of association between two things," *Amer. J. Psychol.*, vol. 15, no. 1, pp. 72–101, 1904.

[6] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, 2008.

[7] R. Řehůřek and P. Sojka, "Software framework for topic modelling with large corpora," in *Proc. LREC Workshop New Challenges for NLP Frameworks*, 2010, pp. 45–50.

Resources

Resource	**Link**
Training corpus — IndicCorpV2-5k	https://huggingface.co/datasets/rony000013/indiccorpV2-5k
Evaluation benchmark — Akhtar et al. (2017)	https://github.com/syedsarfarazakhtar/Word-Similarity-Datasets-for-Indian-Languages
Project source code and artifacts	https://github.com/g25ait2138-collab/word-embeddings-nlp
Colab notebook (Google Drive)	https://colab.research.google.com/drive/102SjYVTS8SguWze8fWloLuwaMKJBQ3Z2